

Name: _____ Partner: _____ Period: _____

Context: Solving Trigonometric Equations

Trigonometric equations arise whenever a periodic model is set equal to a specific output value. In this activity your group will solve trig equations and inequalities in several settings. Choose **one tier** below based on your readiness. Work together — everyone should be ready to explain your group's reasoning.

Tier R — Remediate

The general solution of $\sin \theta = \frac{1}{2}$ is already provided:

$$\theta = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad \theta = \frac{5\pi}{6} + 2\pi k, \quad k \in \mathbb{Z}.$$

1. Substitute $k = 0, 1, 2$ into **Family 1** ($\theta = \frac{\pi}{6} + 2\pi k$) to list three specific solutions.

$k = 0$: $\theta =$ _____ $k = 1$: $\theta =$ _____ $k = 2$: $\theta =$ _____

2. Substitute $k = 0, 1, 2$ into **Family 2** ($\theta = \frac{5\pi}{6} + 2\pi k$) to list three specific solutions.

$k = 0$: $\theta =$ _____ $k = 1$: $\theta =$ _____ $k = 2$: $\theta =$ _____

3. Circle all solutions from problems 1 and 2 that lie in the interval $[0, 4\pi]$.

How many solutions are there in $[0, 4\pi]$? _____

4. The general solution of $\cos \theta = 0$ is: $\theta = \frac{\pi}{2} + \pi k, k \in \mathbb{Z}$.

List all solutions in $[0, 2\pi]$:

$\theta =$ _____

5. In your own words: why does a trig equation have more than one solution in $[0, 4\pi]$?

Tier A — Apply

1. Solve $2\sin\theta + \sqrt{3} = 0$ for all $\theta \in [0, 2\pi)$. Show the four-step procedure from the notes.

Step 1 (isolate): _____ Step 2 (principal): _____ Step 3 (second solution): _____ Step 4 (solutions in $[0, 2\pi)$): _____

2. Solve $2\cos^2\theta - \cos\theta - 1 = 0$ for all $\theta \in [0, 2\pi)$.

Hint: let $u = \cos\theta$ and treat this as a quadratic in u .

Factor: _____ $u =$ _____ or $u =$ _____

Find θ for each value of u : _____

All solutions: $\theta =$ _____

3. **Context:** The depth of water at a pier (in meters) is $d(t) = 2\cos\left(\frac{\pi}{6}t\right) + 4$, $t \in [0, 12]$ hours.

(a) For what values of t is $d(t) = 3$? Show your work. _____

(b) Do any additional solutions from the general formula fall outside $[0, 12]$? Explain why those are excluded.

The table below gives pre-computed values of $f(\theta) = 2 \cos \theta - 1$ at key angles in $[0, 2\pi]$.

θ	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos \theta$	1	$\frac{1}{2}$	0	-1	0	$\frac{1}{2}$	1
$f(\theta)$	1	0	-1	-3	-1	0	1

1. Using the table, identify all $\theta \in [0, 2\pi]$ where $f(\theta) = 0$. These are the *boundary* values for the inequality. $\theta =$ _____

2. On what interval(s) in $[0, 2\pi]$ is $f(\theta) > 0$? Use the table to determine where f is positive, then write your answer as a union of intervals.

3. Verify your answer to #2 algebraically: solve $2 \cos \theta - 1 > 0$ by isolating $\cos \theta$ and applying the general solution structure. _____

4. **Context:** A seasonal daylight model for a city gives $L(t) = 3 \sin\left(\frac{2\pi}{365}(t - 80)\right) + 12$ (hours of daylight), where t is the day of the year, $1 \leq t \leq 365$.

(a) For what days of the year is $L(t) \geq 13.5$? Set up and solve the inequality. Round t to the nearest day. _

(b) Explain in two or three sentences what your answer means in context, and why the general solution (with $+2\pi k$) gives additional values of t that are excluded from this problem. _____
